

$H \in \mathcal{H}$ let Π_H denote that of the two closed subspaces into which H divides $\mathbb{R}^N$ which contains A :

$$A \subset \Pi_H \text{ for } H \in \mathcal{H}. \quad (5.1.23)$$

The thing to prove is that

$$A = \bigcap_{H \in \mathcal{H}} \Pi_H. \quad (5.1.24)$$

By (5.1.23) we have $A \subset \bigcap_{H \in \mathcal{H}} \Pi_H$. Take an $x \notin A$. By Lemma 5.1.11 there exists an $H_0 \in \mathcal{H}$ such that $x \notin \Pi_{H_0}$, whence $x \notin \bigcap_{H \in \mathcal{H}} \Pi_H$. This shows that $\bigcap_{H \in \mathcal{H}} \Pi_H \subset A$. Hence (5.1.24) follows. $\square$

5.2 Additive functions

A function⁶ $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is called *additive* iff it satisfies Cauchy's functional equation

$$f(x + y) = f(x) + f(y) \quad (5.2.1)$$

for all $x, y \in \mathbb{R}^N$. For $N = 1$ equation (5.2.1) was first treated by A. M. Legendre [206] and C. F. Gauss [96], but A. L. Cauchy [41] first found its general continuous solution, and the equation has been named after him. Concerning equation (5.2.1) and the vast literature of the subject, cf. also Aczél [5].

It follows from (5.2.1) by induction that if $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is additive, then

$$f\left(\sum_{i=1}^n x_i\right) = \sum_{i=1}^n f(x_i) \quad (5.2.2)$$

for every $n \in \mathbb{N}$ and for every $x_1, \dots, x_n \in \mathbb{R}^N$.

Lemma 5.2.1. *If $f_1 : \mathbb{R}^N \rightarrow \mathbb{R}$ and $f_2 : \mathbb{R}^N \rightarrow \mathbb{R}$ are additive functions, then, for every $a, b \in \mathbb{R}$, the function $f = af_1 + bf_2$ is additive.*

This results immediately from (5.2.1).

The following theorem gives a fundamental property of solutions of equation (5.2.1).

Theorem 5.2.1. *If $f : \mathbb{R}^N \rightarrow \mathbb{R}$ satisfies equation (5.2.1), then*

$$f(\lambda x) = \lambda f(x) \quad (5.2.3)$$

for every $x \in \mathbb{R}^N$ and $\lambda \in \mathbb{Q}$.

Proof. For $x = y = 0$ we get from (5.2.1)

$$f(0) = 0, \quad (5.2.4)$$

⁶ So in the present book we consider only finite-valued additive functions. Additive functions with infinite values have been considered by I. Halperin [131]. He showed that the only such functions are $f = +\infty$ and $f = -\infty$ (cf. also 16.6).

whence, setting in (5.2.1) $y = -x$, we obtain

$$0 = f(0) = f(x - x) = f(x) + f(-x),$$

i.e.,

$$f(-x) = -f(x). \quad (5.2.5)$$

Thus the function f is odd.

Taking in (5.2.2) $x_1 = \dots = x_n = x$ we obtain (5.2.3) for all $\lambda \in \mathbb{N}$. According to (5.2.4) and (5.2.5), (5.2.3) holds for all $\lambda \in \mathbb{Z}$.

Now, an arbitrary $\lambda \in \mathbb{Q}$ may be written as $\lambda = \frac{k}{m}$, where $k \in \mathbb{Z}, m \in \mathbb{N}$. Hence $kx = m(\lambda x)$ and so, by what already has been proved,

$$kf(x) = f(kx) = f(m(\lambda x)) = mf(\lambda x),$$

whence (5.2.3) follows. $\square$

If $N = 1$, then we get from (5.2.3) with $c = f(1)$

$$f(\lambda x) = c\lambda \text{ for } \lambda \in \mathbb{Q},$$

whence, if f is continuous,

$$f(x) = cx \text{ for } \lambda \in \mathbb{R}. \quad (5.2.6)$$

Formula (5.2.6) gives the general continuous additive functions $f : \mathbb{R} \rightarrow \mathbb{R}$. For continuous additive functions $f : \mathbb{R}^N \rightarrow \mathbb{R}$, $N > 1$, the corresponding formula will be found in 5.5.

Theorem 5.2.1 says that every additive function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ is a homomorphism from the space $(\mathbb{R}^N; \mathbb{Q}; +; \cdot)$ into $(\mathbb{R}; \mathbb{Q}; +; \cdot)$ (cf. 4.3). Thus as an immediate consequence of Theorem 4.3.1 we obtain the following

Theorem 5.2.2. *Let H be an arbitrary Hamel basis of the space $(\mathbb{R}^N; \mathbb{Q}; +; \cdot)$. Then for every function $g : H \rightarrow \mathbb{R}$ there exists a unique additive function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ such that $f|_H = g$.*

Theorem 5.2.2 gives the general construction of all the additive functions $f : \mathbb{R}^N \rightarrow \mathbb{R}$. In fact, every additive f may be obtained as the unique additive extension of a certain function $g : H \rightarrow \mathbb{R}$, viz. $g = f|_H$.

For many years the existence of discontinuous additive functions was an open problem. Mathematicians could neither prove that every additive function is continuous (which is false), nor exhibit an example of a discontinuous additive function (as we will see later, it was a hopeless task, since there do not exist effective examples of discontinuous additive functions). It was G. Hamel who first succeeded in proving that there exist discontinuous additive functions. In Hamel [134] he proved Theorem 5.2.2 above (for $N = 1$), whence the existence of discontinuous additive functions follows easily. In fact, we obtain from Theorem 5.2.2

Corollary 5.2.1. *Let H be an arbitrary Hamel basis of the space $(\mathbb{R}; \mathbb{Q}; +; \cdot)$, and let $g : H \rightarrow \mathbb{R}$ be an arbitrary function. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the unique additive extension of g . The function f is continuous if and only if $g(x)/x = \text{const}$ for $x \in H$.*

Proof. If f is continuous, then it has form (5.2.6), and in particular, for $x \in H$, we must have $g(x) = f(x) = cx$, i.e., $g(x)/x = c = \text{const}$. If $g(x) = cx$ for $x \in H$, then the function cx is an additive extension of g , and by the uniqueness of such an extension f must be given by (5.2.6). Thus f is continuous. $\square$

Now, it is enough to take any function $g : H \rightarrow \mathbb{R}$ such that $g(x)/x \neq \text{const}$ on H , and as an additive extension of g we obtain a discontinuous additive function.

This argument would have to be modified for $N > 1$, but we can also argue otherwise. It follows from the proof of Theorem 4.3.1 that the values of an additive extension $f : \mathbb{R}^N \rightarrow \mathbb{R}$ of a function $g : H \rightarrow \mathbb{R}$ are given as linear combinations of the values of g (at suitable points of H) with rational coefficients. Hence, if g takes on only rational values, then also all values of f will be rational, and thus f cannot be continuous. We shall often make use of this observation, therefore we will formulate it as

Corollary 5.2.2. *Let H be an arbitrary Hamel basis of the space $(\mathbb{R}^N; \mathbb{Q}; +; \cdot)$, and let $g : H \rightarrow \mathbb{R}$ be a function with rational values only ($g(H) \subset \mathbb{Q}$), $g \neq 0$. Then the additive extension $f : \mathbb{R}^N \rightarrow \mathbb{R}$ of g is discontinuous.*

Hence we infer the existence of discontinuous additive functions $f : \mathbb{R}^N \rightarrow \mathbb{R}$ for arbitrary $N \in \mathbb{N}$.

Discontinuous additive functions are sometimes called *Hamel functions*. They exhibit many pathological properties, as will be seen later in this book. Since such functions are undesirable in applications, we will try to find possibly weak conditions which assure the continuity of an additive function.

5.3 Convex functions

Let $D \subset \mathbb{R}^N$ be a convex and open⁷ set. A function $f : D \rightarrow \mathbb{R}$ is called *convex*⁸ iff it satisfies Jensen's functional inequality

$$f\left(\frac{x+y}{2}\right) \leq \frac{f(x) + f(y)}{2} \quad (5.3.1)$$

for all $x, y \in D$. If the inequality in (5.3.1) for $x \neq y$ is sharp, f is called *strictly convex*.

If a function $f : D \rightarrow \mathbb{R}$ (where $D \subset \mathbb{R}^N$ is a convex and open set) satisfies the inequality

$$f\left(\frac{x+y}{2}\right) \geq \frac{f(x) + f(y)}{2} \quad (5.3.2)$$

⁷ In this book we consider convex functions only on convex and open subsets of $\mathbb{R}^N$, and we consider only finite-valued functions. Continuous convex functions defined on subsets of $\mathbb{R}^N$ which are convex, but not necessarily open, and also infinite-valued convex functions, are discussed in detail in Rockafellar [268]. Adjoining to D also some of its frontier points would seriously complicate the presentation, but the essentials would remain the same. (Let us note also that convex functions are often considered in more general spaces than $\mathbb{R}^N$; cf., e.g., Roberts–Varberg [267]).

⁸ Some authors call functions fulfilling (5.3.1) *J-convex* or *Jensen-convex*, reserving the name *convex* for functions fulfilling (5.3.2) for all real $\lambda \in [0, 1]$. Since in the present book we are primarily interested in functions fulfilling Jensen's inequality (5.3.1), we call such functions simply *convex*, as many authors do. Functions fulfilling (5.3.2) for all real $\lambda \in [0, 1]$ will be referred to as *continuous and convex*. As we shall see later, they are necessarily continuous.